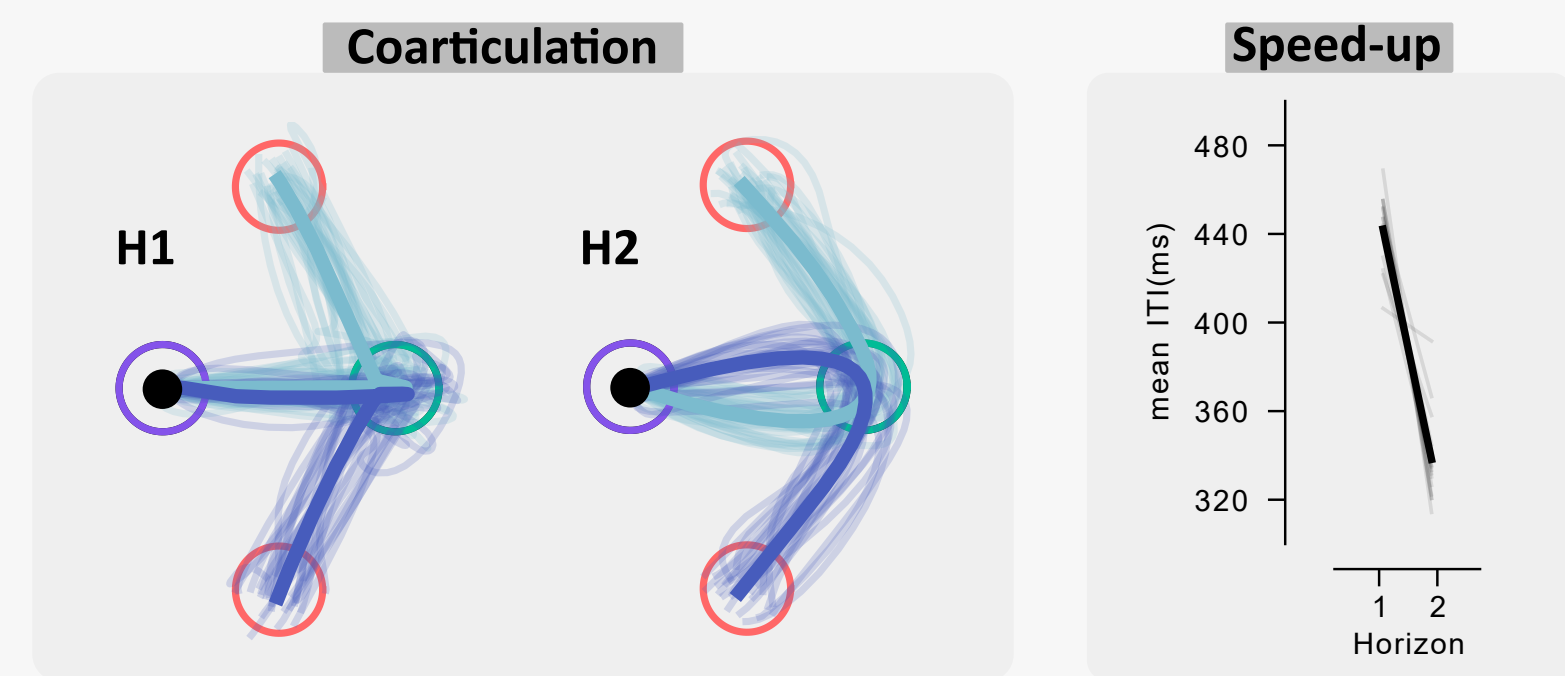




Background

Action sequences are performed with striking fluency:

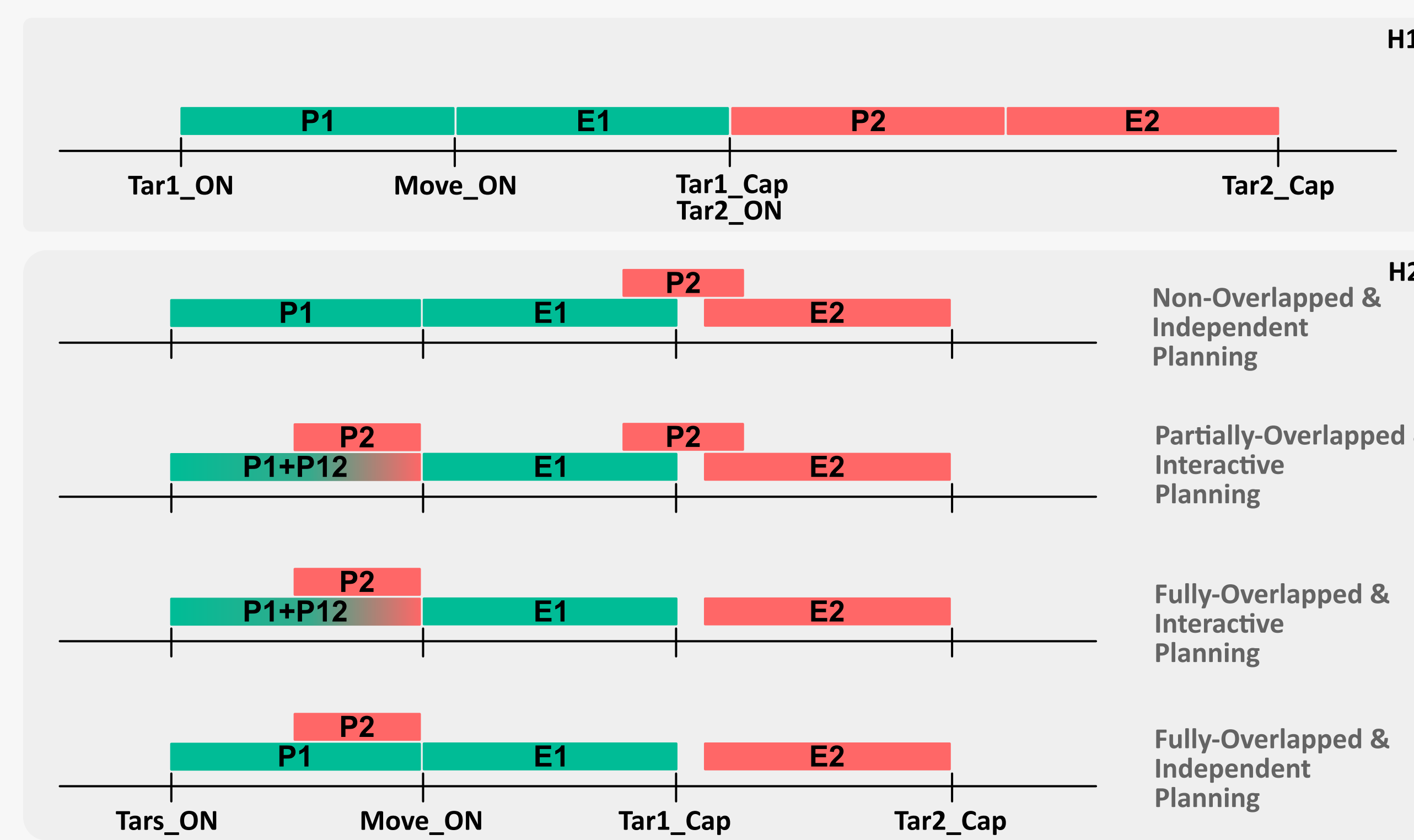


Here, we investigate how the **cortical motor regions** supports the preparation and execution of coarticulated reaching sequences.

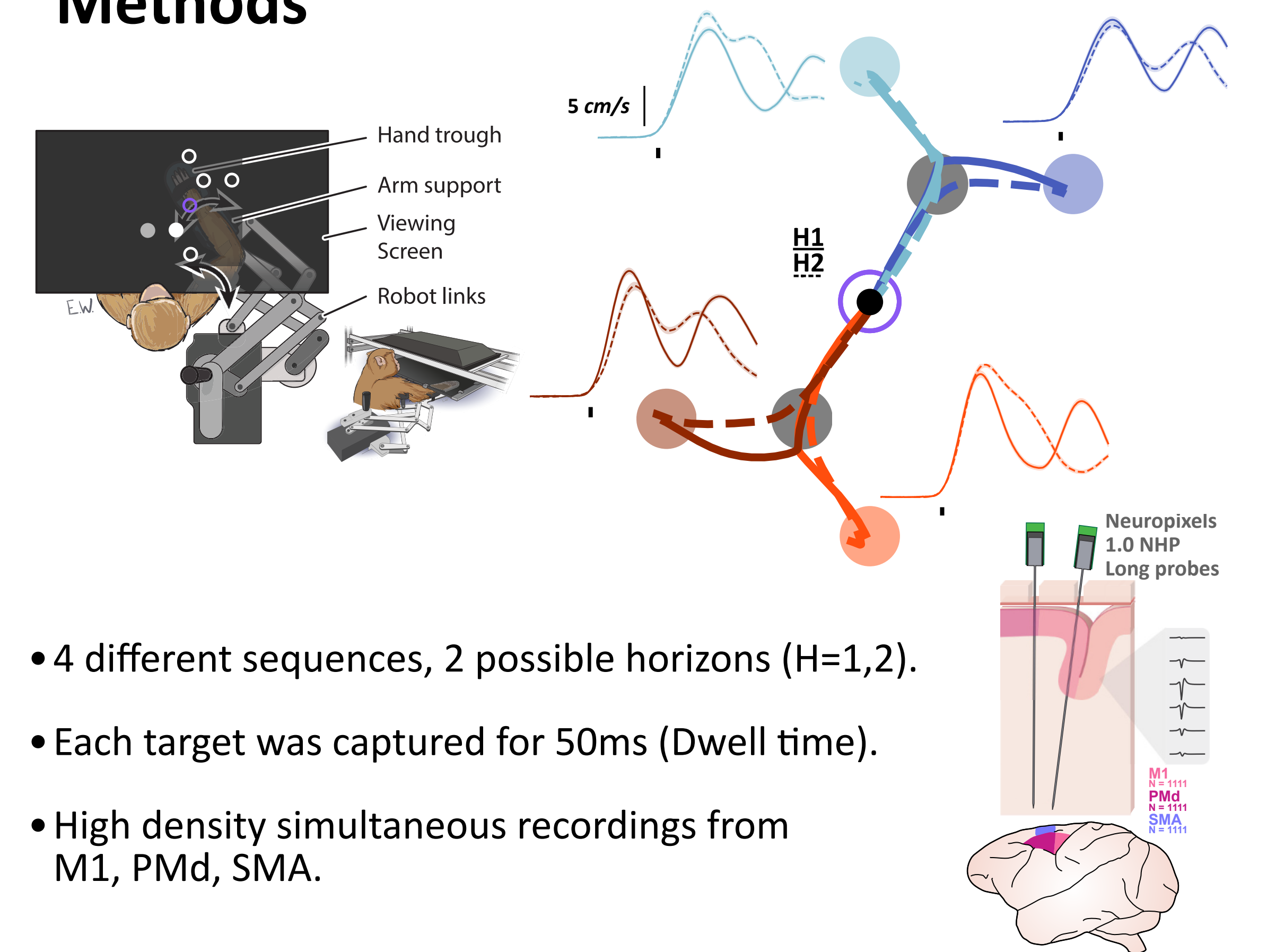
We specifically ask:

Whether the planning of the two reaches **temporally overlap**, and if so, whether they are **independent or interactive**?

Potential Mechanisms

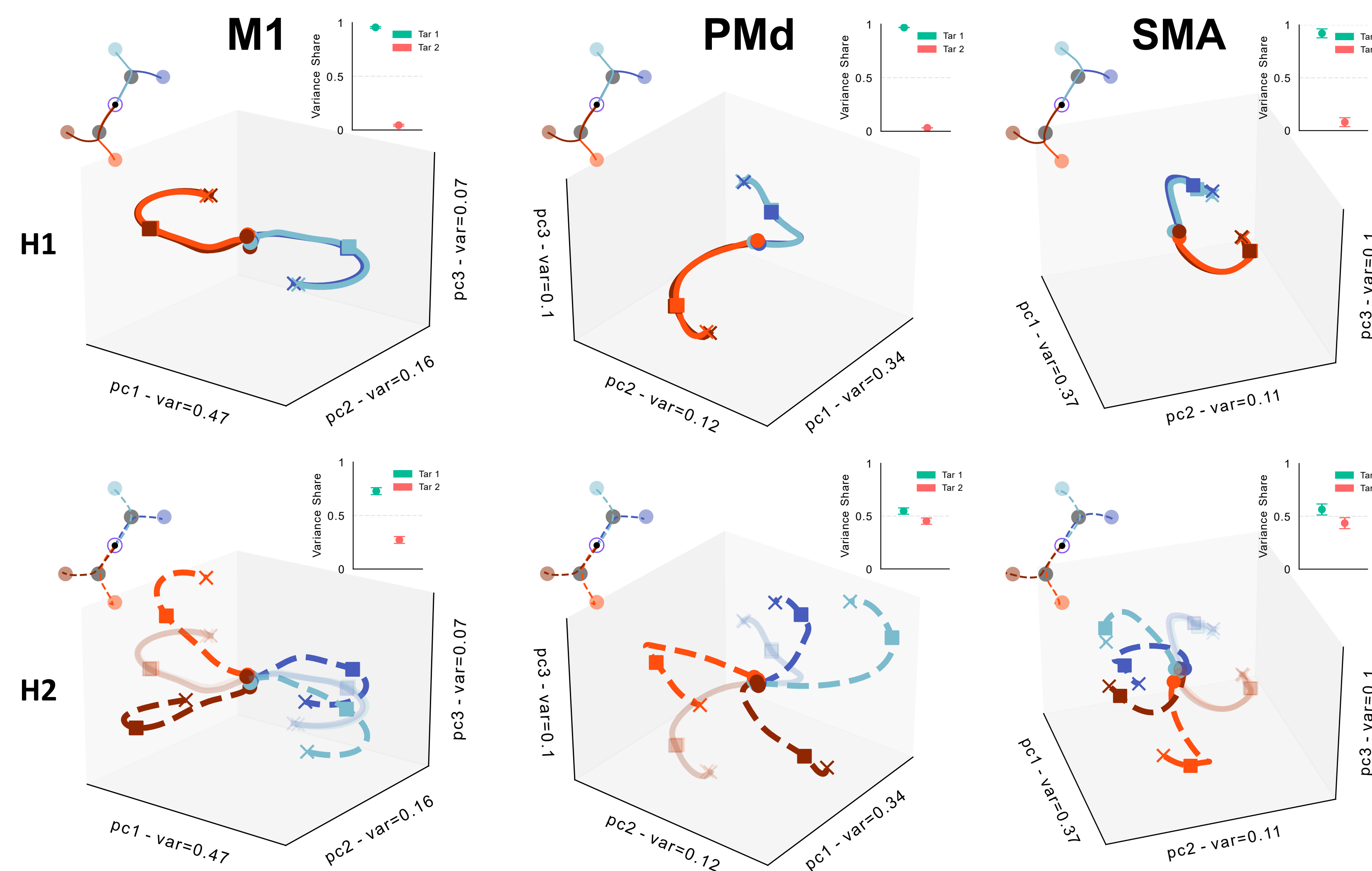


Methods

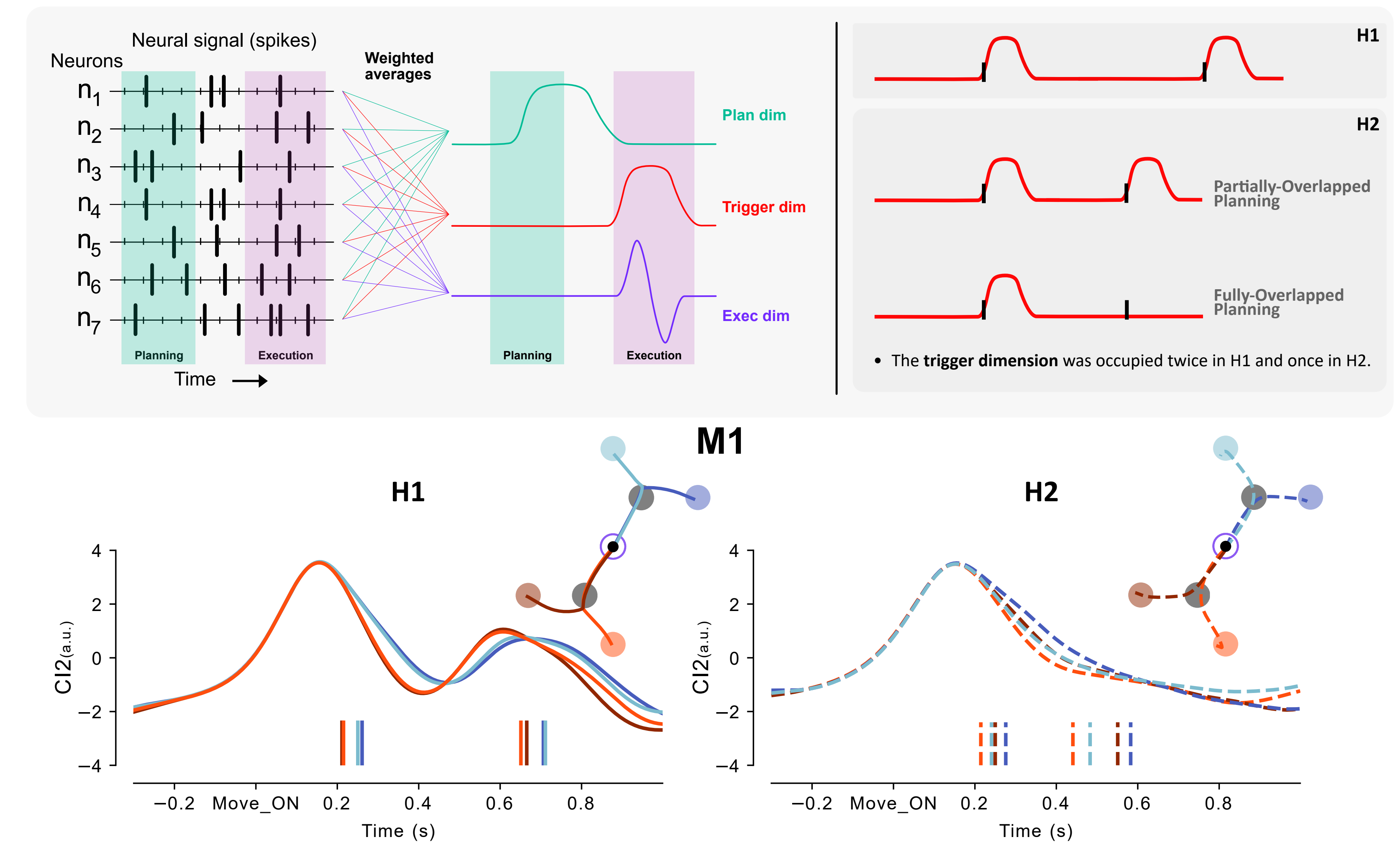


- 4 different sequences, 2 possible horizons (H=1,2).
- Each target was captured for 50ms (Dwell time).
- High density simultaneous recordings from M1, PMd, SMA.

Sequence specific representations pre- and during movement



Fully-Overlapped planning of the sequence



References

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- Ames, K.C., Ryu, S.I. & Shenoy, K.V. Simultaneous motor preparation and execution in a last-moment reach correction task. *Nat Commun*
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- Wang, T., Chen, Y., Zhang, Y. et al. Multiplicative joint coding in preparatory activity for reaching sequence in macaque motor cortex. *Nat Commun*
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Summary

- The planning of the two reaches in the sequence **fully overlap temporally** before the movement onset in M1, PMd, and SMA.
- The planning of the two reaches are **not fully independent** from each other.

Discussion

- How aligned P1 and P2 are? Are they fully overlapping? or nearly orthogonal? How does this alignment change across areas?
- How differently the sequences are represented across areas?
- What will happen when facing longer sequences?
- How do these areas orchestrate together to produce sequences?